

EXERCISE 1

This is the basic LAN(Local Area Network) setup. It includes basic IP addressing, ping testing to test if the 2 pcs are able to communicate with each other. The devices are connected using a networking device called a switch.

Goal: Learn basic IP addressing(Using the GUI) and ping testing.

Devices:

- 2 PCs
- 1 Switch
- Copper straight-through cables (ETHERNET CABLES)

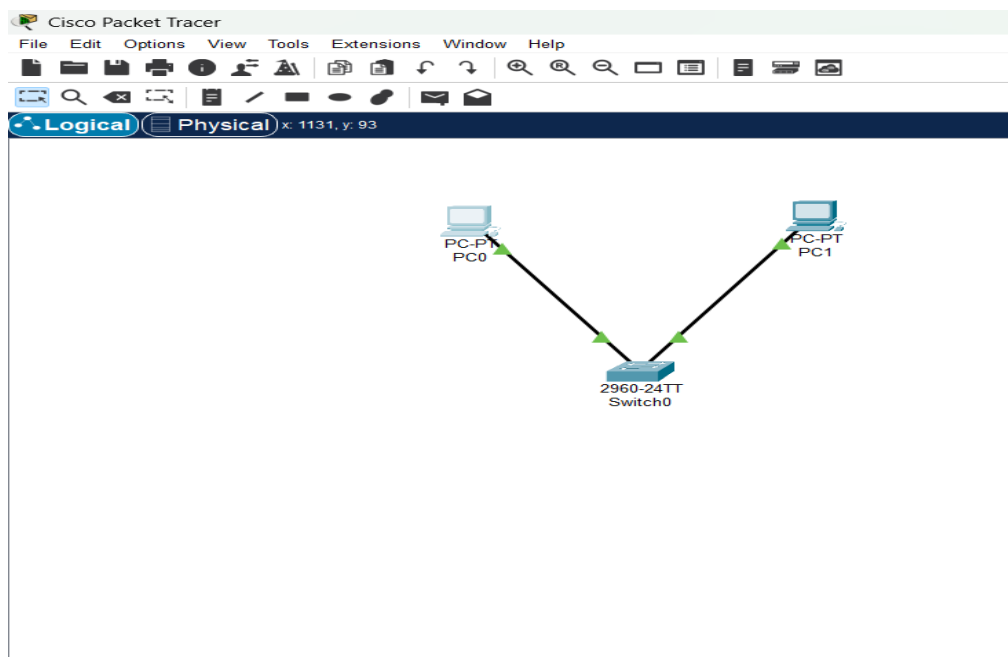
Tasks:

1. Connect both PCs to the switch.
2. Assign IP addresses:

Device	IP Address	Subnet Mask	Default Gateway
PC0	192.168.1.1	255.255.255.0	N/A
PC1	192.168.1.2	255.255.255.0	N/A

3. Ping PC1 from PC0.

HERE IS AN IMAGE OF THE NETWORK



What is a LAN?

LAN stands for local area network. This is a network that is a local network, and it is formed when you connect devices together using a network device called a switch. This is a local network so its not connected to the internet at the moment, to be able to connect with other networks there is another device that you will need and that's called a router which is not in this project.

How do switches connect devices?

Your pcs and the switch have ports, they are fast ethernet ports(FA0/X). These ports are then used to connect devices together using cables. The cables used in this project are copper straight through cables which are ethernet cables in real world scenarios. Once the pcs are connected to the switch, they are now in a LAN.

How IP addresses work

These are unique identifiers that are assigned to your device manually or automatically. They are like your PCs phone number that can be used to communicate with other devices.in this project they are in the same network so they will be able to communicate

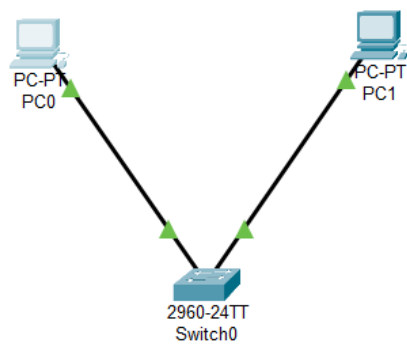
How to test connectivity with ping

Ping is a command that is used to test connectivity between device in a network. Its very simple to use. To test connectivity, you just have to know the ip of the device you want to know is connected.

Syntax: ping ip address

Eg.1 pinging from pc0 to pc1

Ping 192.168.1.2. (look at results below)



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PC0
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=17ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 17ms, Average = 4ms

C:\>
```

The screenshot shows the 'Command Prompt' window of PC0. It displays the output of a 'ping 192.168.1.1' command. The output shows four successful replies from 192.168.1.1, each with 32 bytes of data and a time of less than 1ms. The ping statistics at the bottom confirm that 4 packets were sent and received, with 0% loss.

When it says 4 packets have been received it means that connection has been established and the devices can communicate. The packets are ICMP packets(internet control message protocol). They are like “ARE YOU THERE” to the target device to test whether its online. As you can see there is a reply from the target device that’s acts like its saying “YES IM HERE”, it’s an acknowledgement of connection.

In conclusion, this exercise showed how to create a simple LAN using two PCs and a switch. By assigning IP addresses in the same subnet and testing with the ping command, I confirmed that both PCs could communicate successfully.